

Building AI Agents in Enterprise

Week 1 - Lecture 1

Abhijith Neerkaje & Ajay Shenoy

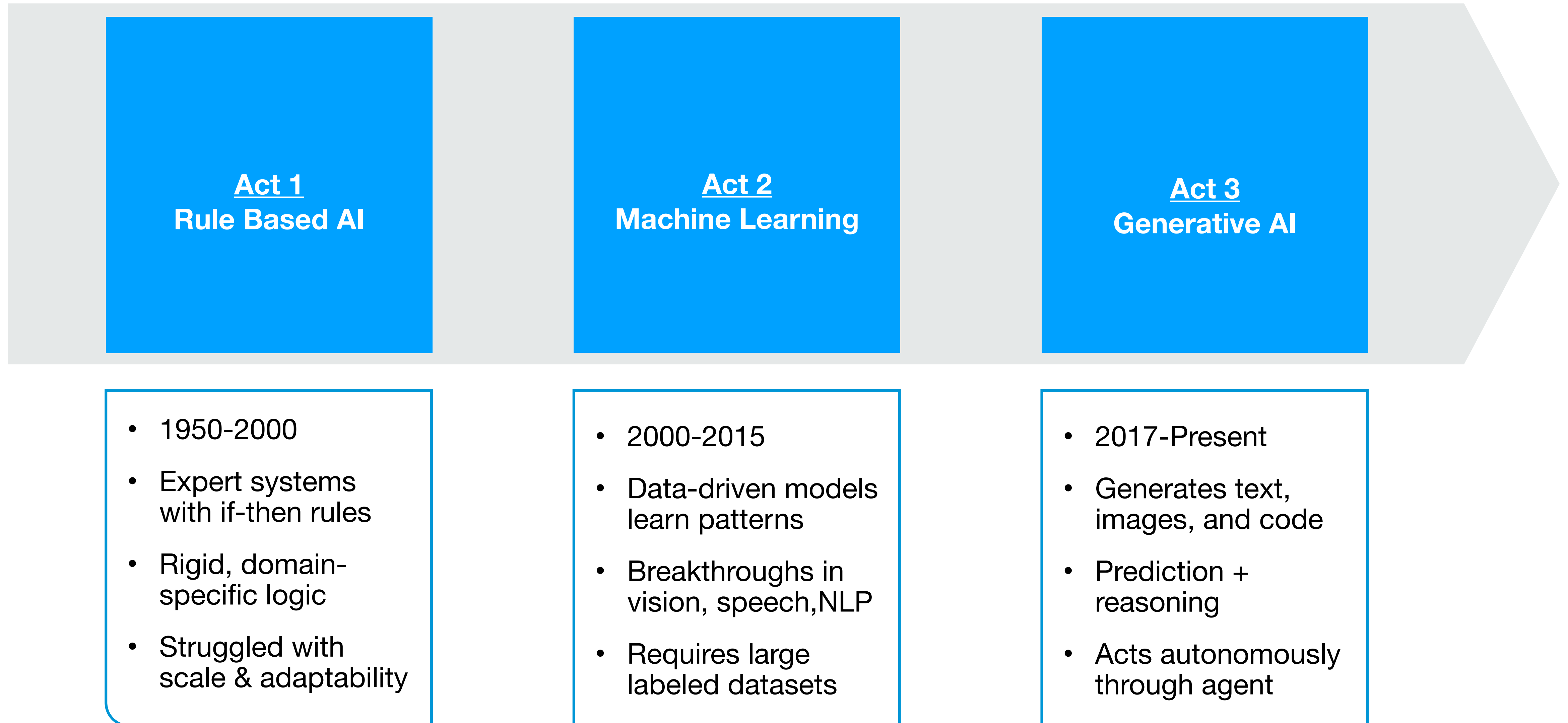
Course Logistics and plan for the next seven weeks

Week	Class	Date	Topic
1	1	14/2	Gen AI Fundamentals
	2	15/2	Prompt Engineering , Building Agents
2	3	21/2	RAG Fundamentals
	4	22/2	Design Choices in RAG
3		28/2	MID COURSE BREAK
		1/3	
4	5	7/3	Tools & MCP
	6	8/3	Memory in Agents
5	7	14/3	Evaluation of Agents
	8	15/3	LLM as a Judge / Agentic RAG
6	9	21/3	Inference Optimization ,Caching and Fine Tuning Models
	10	22/3	Guardrails and Observability
7	11	28/3	Deploying agents
	12	29/3	Multi Agent Architectures and Wrap up

Plan for today

- Gen AI and Agentic AI in historical context
- What is an AI Agent?
- Foundation Models - Brains of an Agent
- Tools - Body and senses of an Agent
- Thought Action Observe Cycle of Agents
- Application of agents in Enterprise
 - Opportunities
 - Challenges
- AI Technology Stack

How did we arrive here?



What is an AI Agent?

- **Definition:** AI agent is as an software application that attempts to **achieve a goal** by **observing** the world and **acting** upon it using the **tools** that it has at its disposal.

Reasoning

- Use logic, memory, and information to infer, conclude, and solve problems

Acting

- Execute tasks or actions based on decisions or inputs

Observing

- Perceiving and gathering information from the environment to understand context

Planning

- Devise strategies and sequence actions to achieve defined goals.

Collaborating

- Work with humans or other agents through communication & coordination

Self-Refining

- Learn, adapt, and improve its performance through feedback and experience.

Types of agentic systems

Autonomy

- Workflow Agent
- Semi Autonomous Agent
- Autonomous Agent

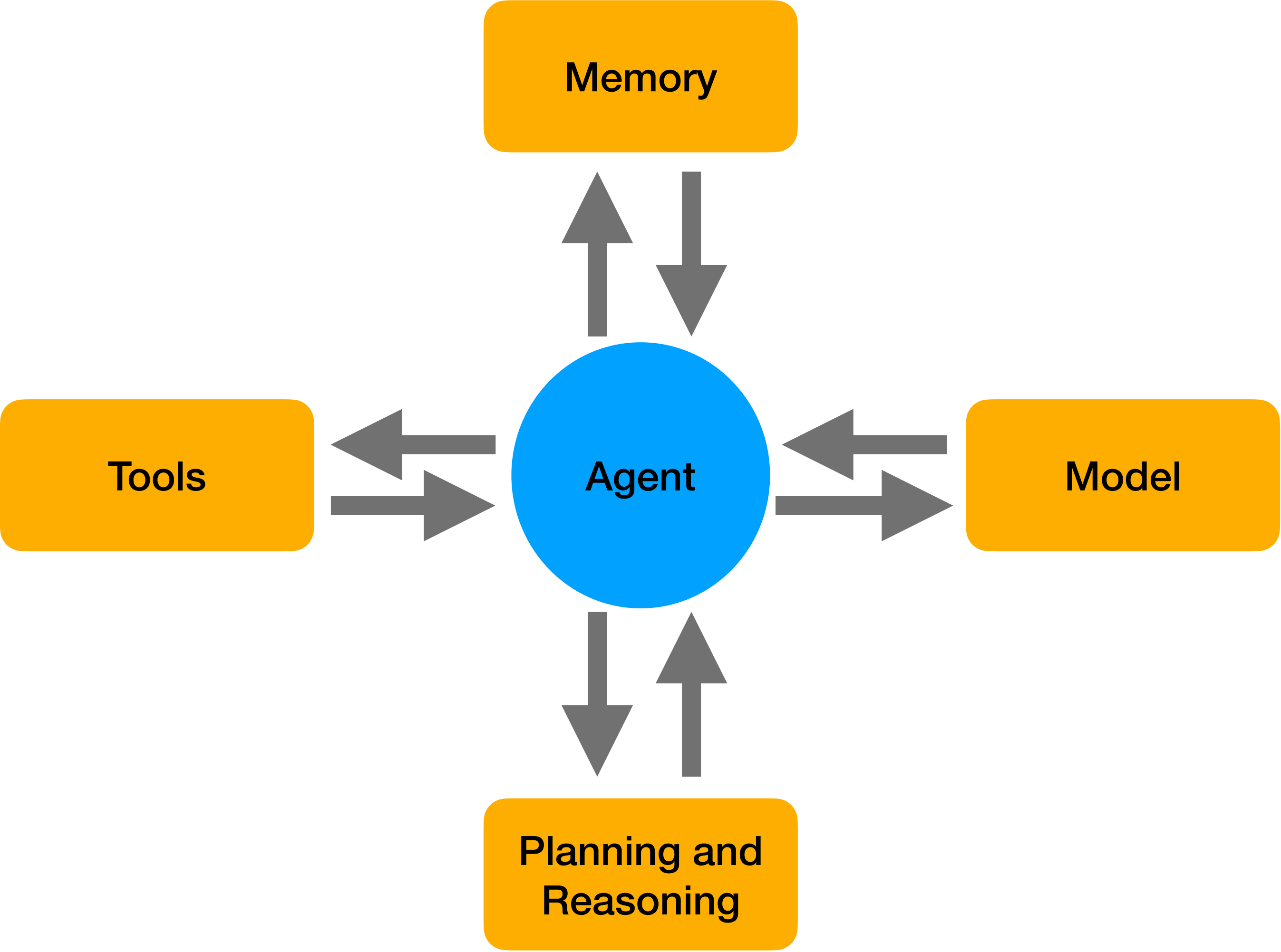
Interaction Mode

- Interactive partners / Surface agents
- Background agents

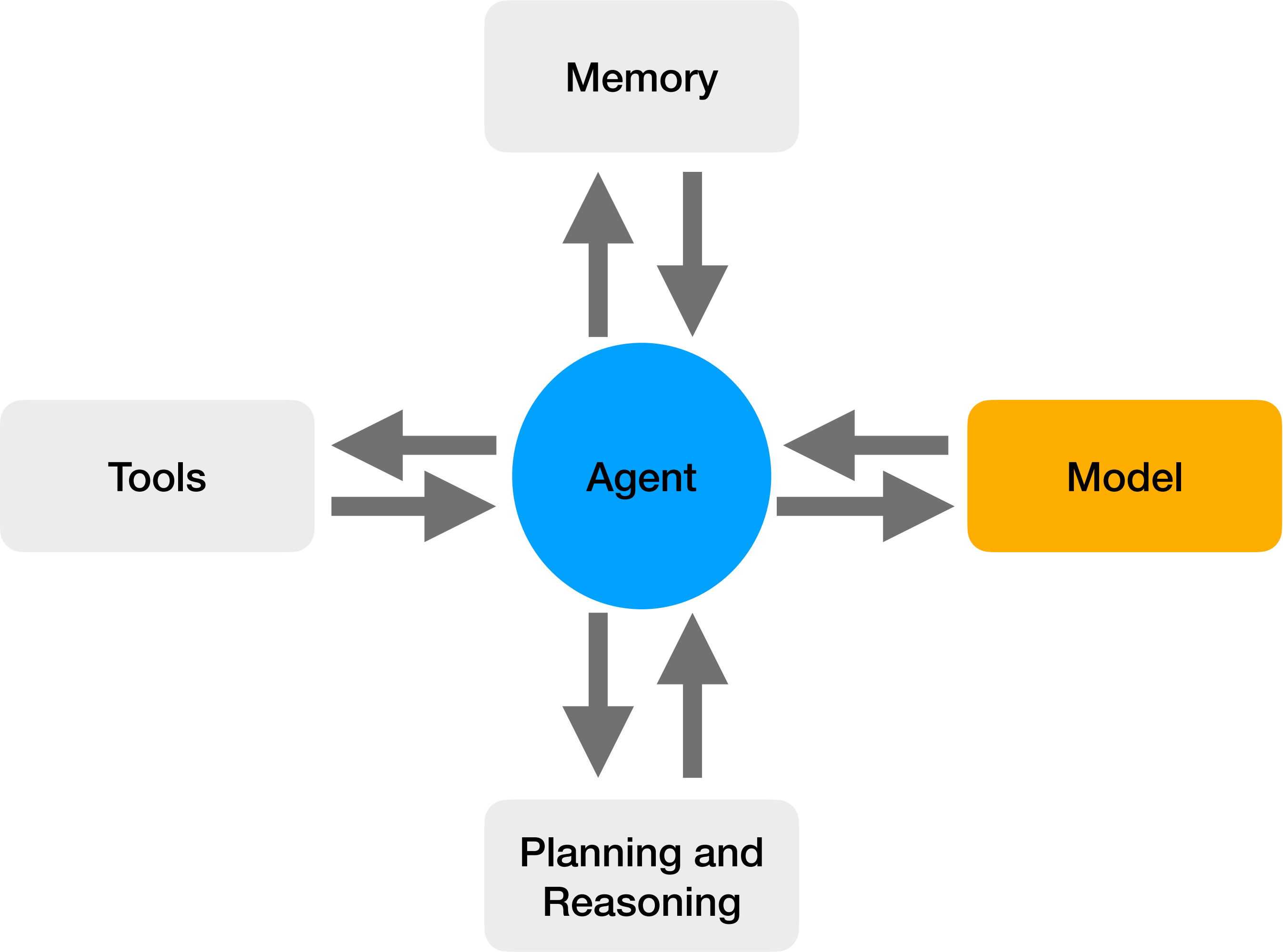
Number of Agents

- Single agent system
- Multi agent system

Components of an agentic systems



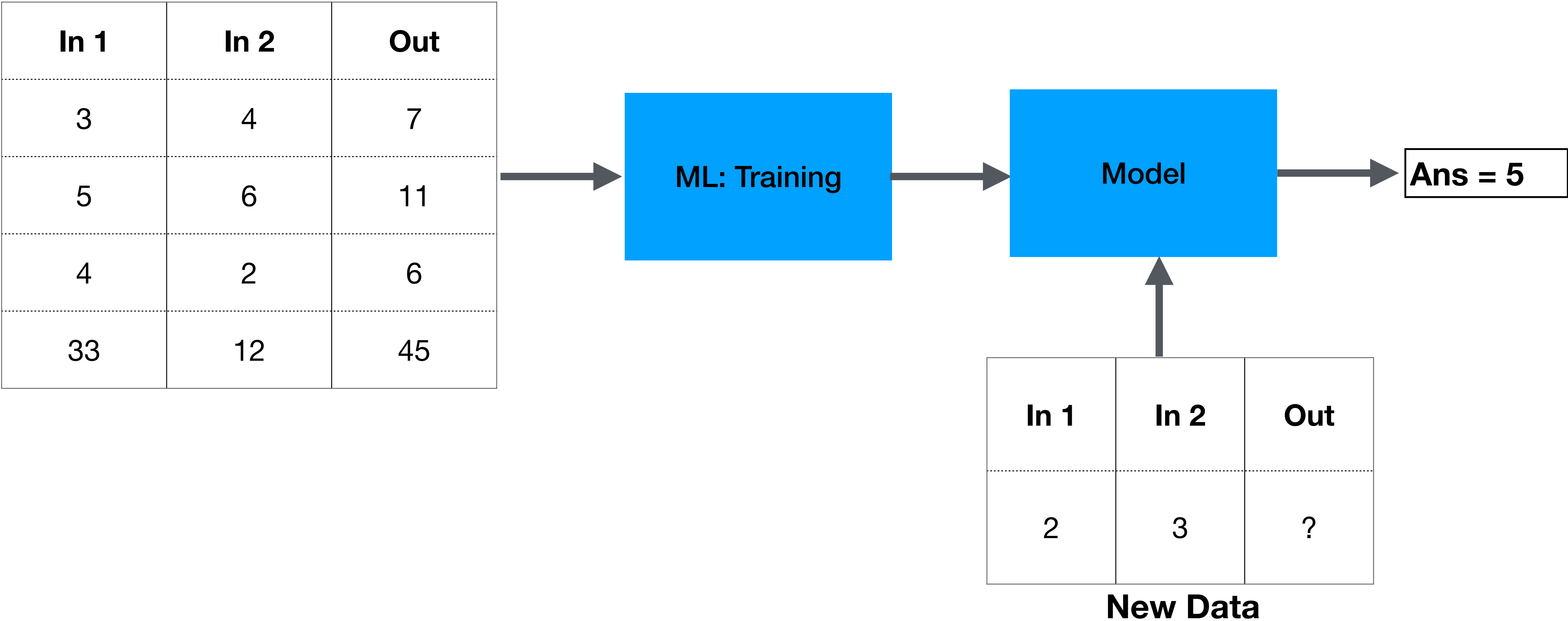
Let us first focus on the model



What is a model?

An **AI model** is a computer program that has been trained on a large amount of data to recognize patterns, make predictions, classify content or generate content. Learn from **examples** rather than explicit programming.

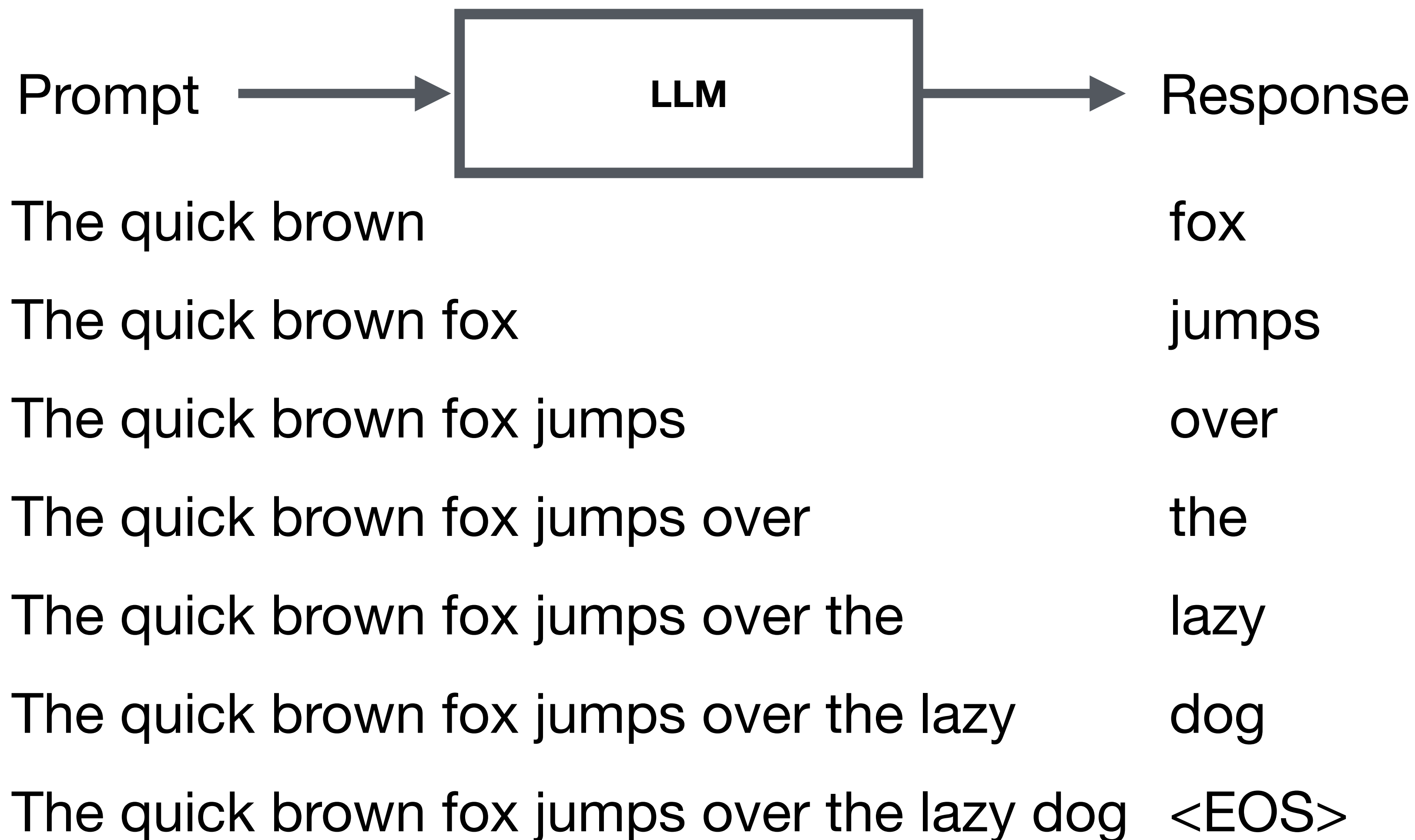
Examples



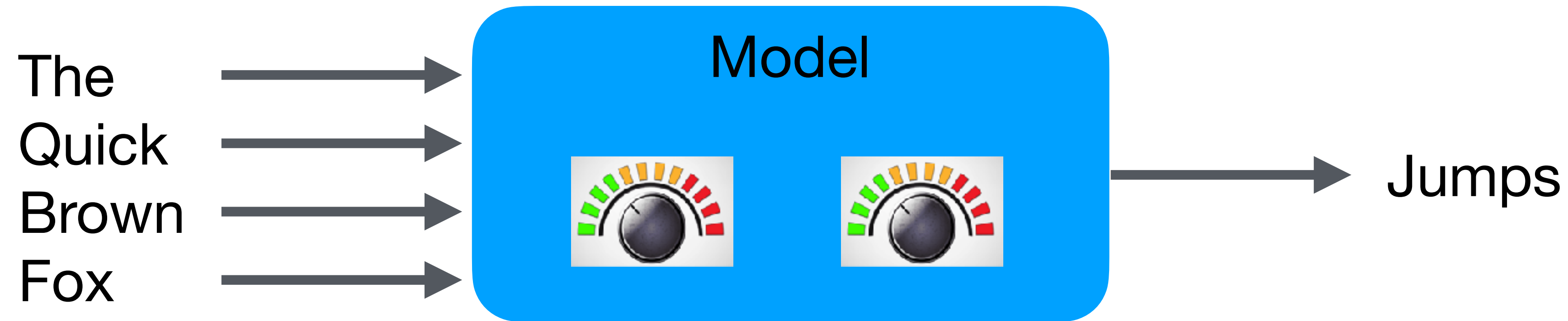
Language Model

- A language model is an AI system trained to understand and generate human-like text.
- **Predicts** the next word or token based on context.
- A **token** is a word a part of a word or a punctuation. They are the currency of LLM applications
- **Context length** is the number of tokens the model can handle at a time
- Language models power applications like chatbots, translation, summarization, and code generation

How does a language model generate this response?



What is “Large” about the Large Language Model?

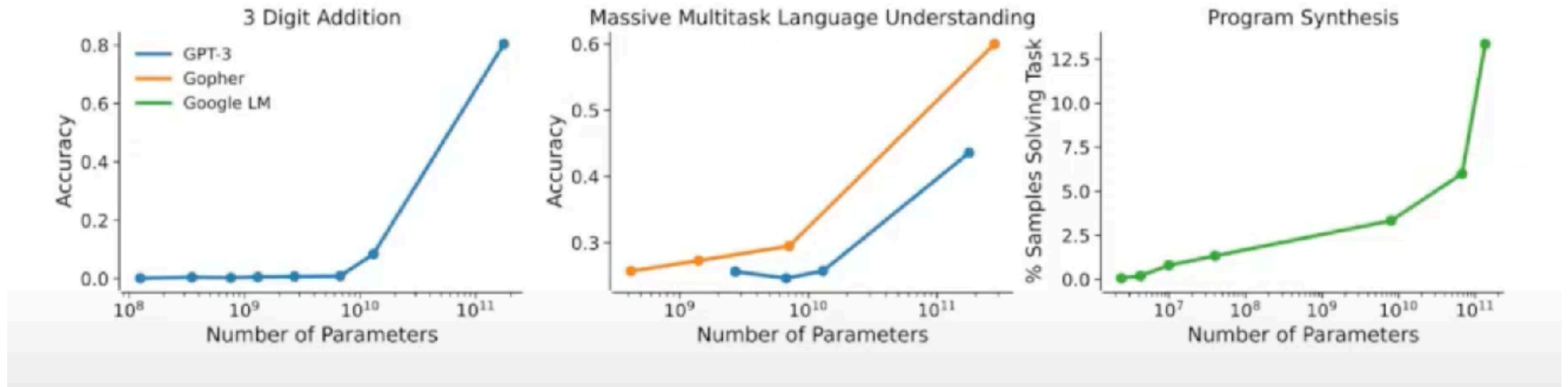


- Think of models as boxes with tunable knobs. You can tweak the knobs to get desired output. Each knob that can be tweaked is a **parameter**
- Large language models (LLMs) have large number of (**billions and trillions**) parameters
- You tweak the parameters based on data. Large language models are trained on large corpus (**internet scale**) of data

How large are they after all?

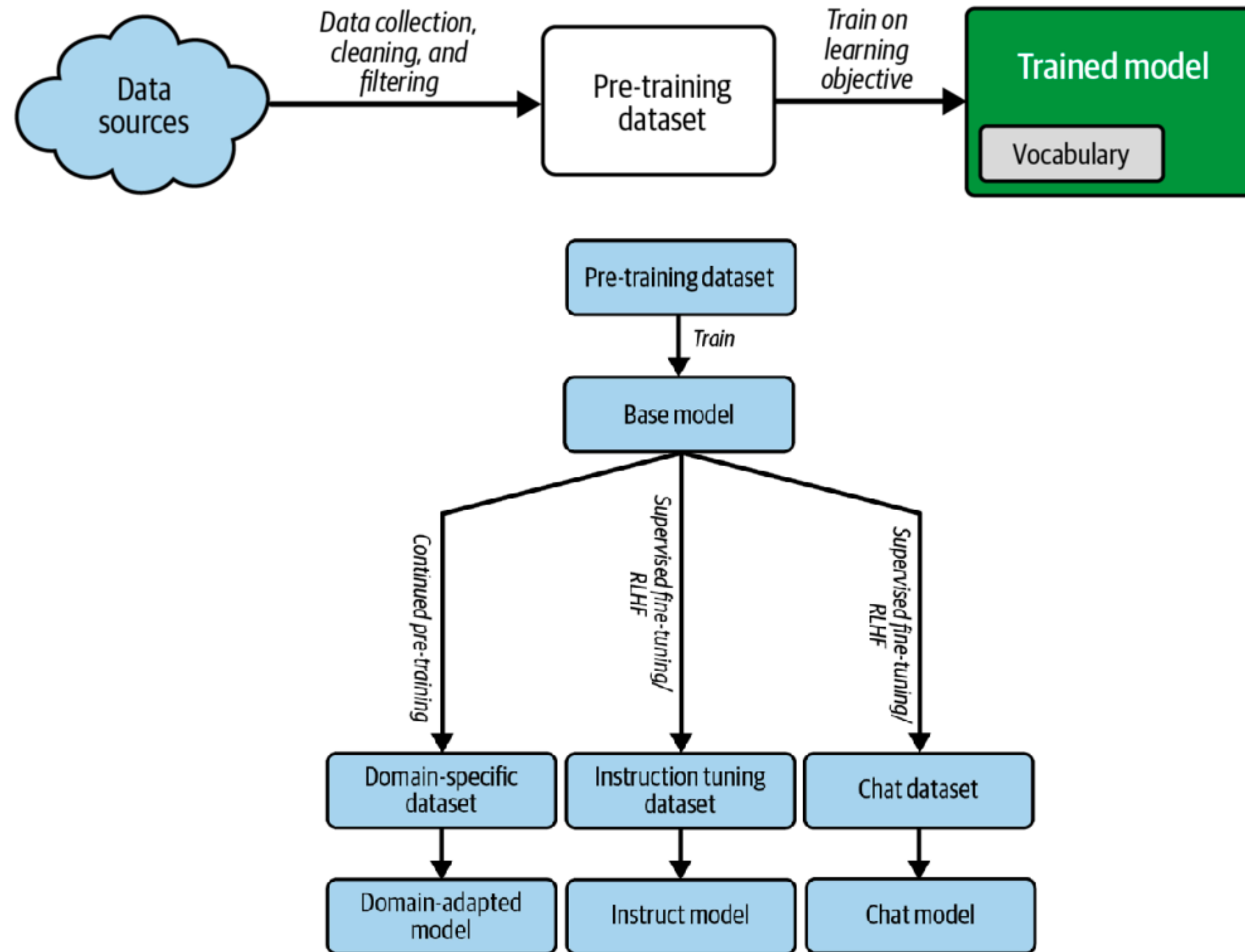


Why should LLMs be large?



Model performance jumps abruptly at certain size

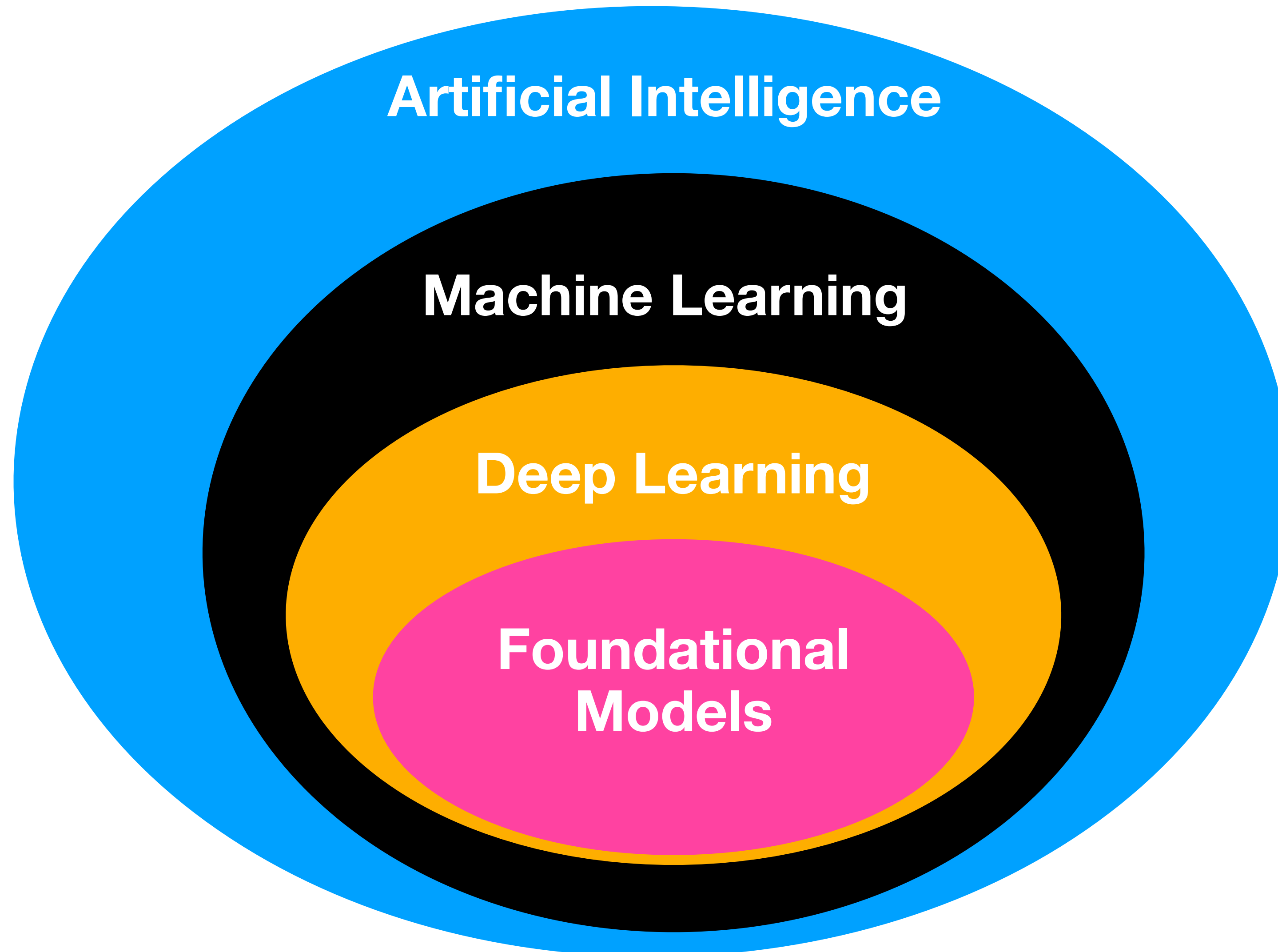
How are large language models built?



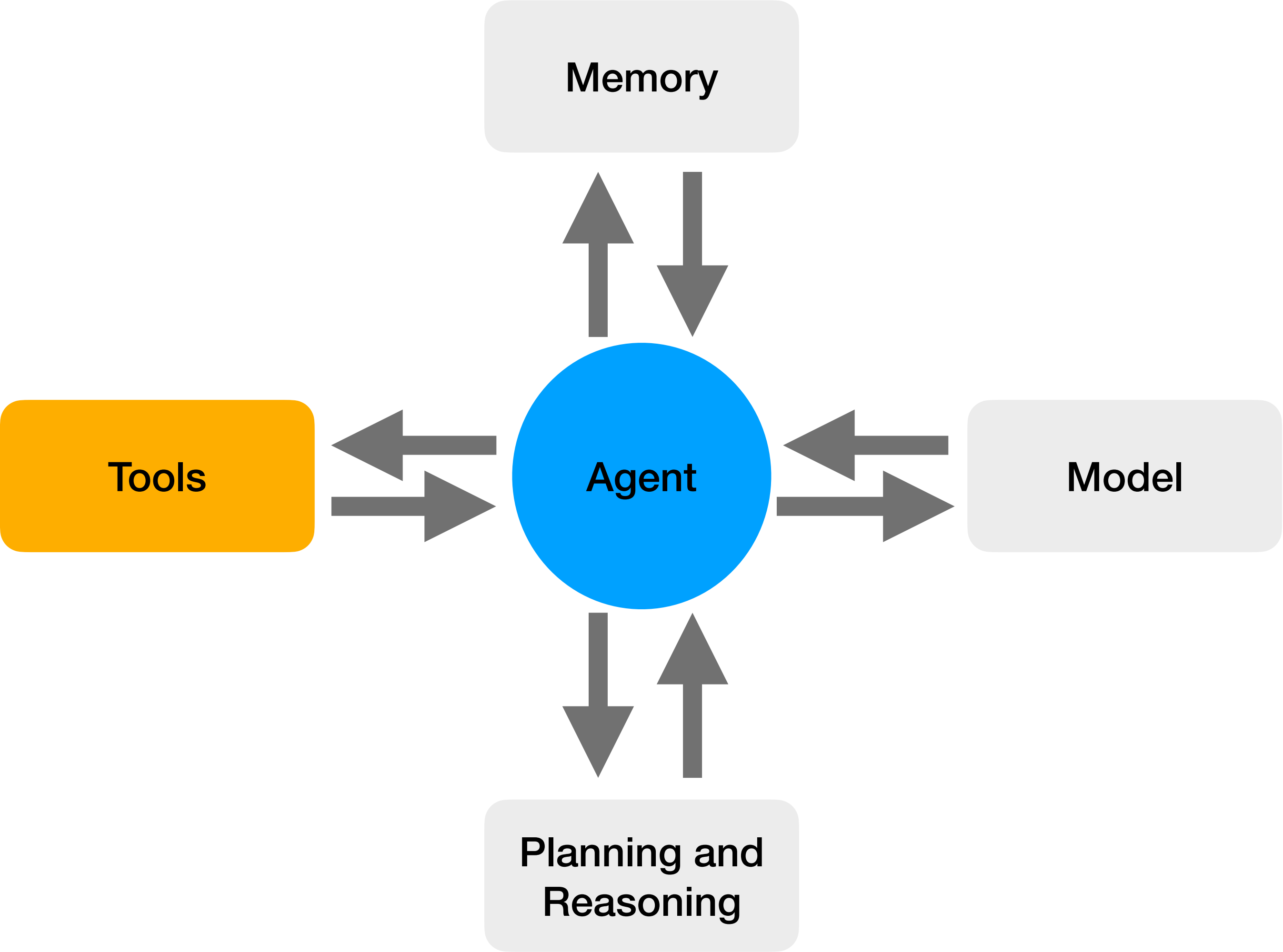
Other types of models also exist

Model Type	Input → Output	Example
Language Models (LMs)	Text → Text	GPT-4 generating a marketing email from bullet points.
Reasoning Models	Structured/Unstructured Data → Decision/Action	Deep Seek r1 , o series of models from open AI
Text-to-Speech (TTS) Models	Text → Speech	Amazon Polly reading out product descriptions.
Speech-to-Speech (S2S) Models <i>(Voice-based models)</i>	Speech → Speech	Meta’s SeamlessM4T translating spoken English to spoken Spanish instantly.
Text-to-Image (T2I) Models	Text → Image	DALL·E generating a product mock-up from a design brief.
Text-to-Video (T2V) Models	Text → Video	OpenAI Sora creating a short ad video from a written script.
Multimodal models	Any modes → Any mode	GPT-4o

How are foundational models related to AI, ML, DL etc?



Let us now focus on the tools



What is a Tool?

Definition: Tools are external functions, APIs, or systems that an AI agent can invoke to gather information, perform actions, or interact with the environment.

Knowledge & Data Tools

- Databases
- Vector Stores

Computation Tools

- Optimizers
- Calculators

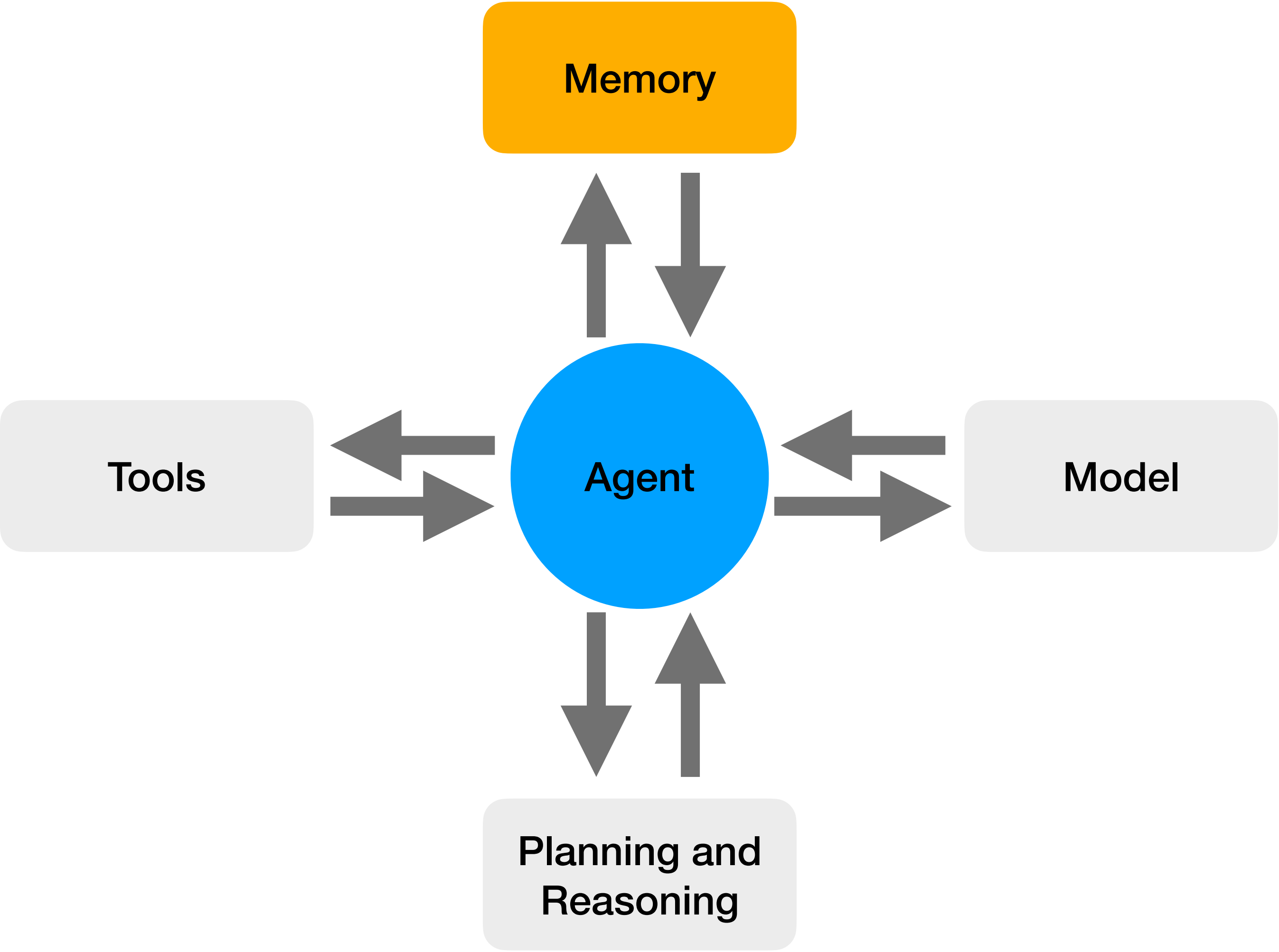
Communication Tools

- Email service
- Slack / WhatsApp or CRM connectors

External Service APIs

- Payment Gateway
- Weather API

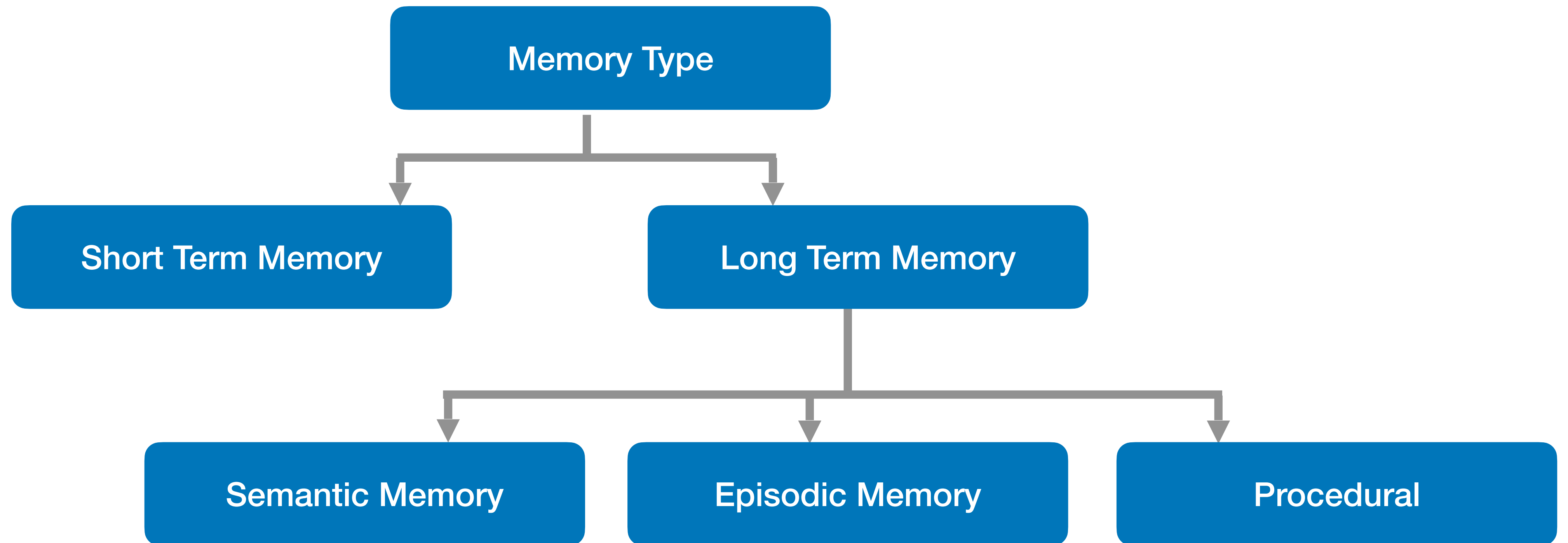
Let us now focus on the memory



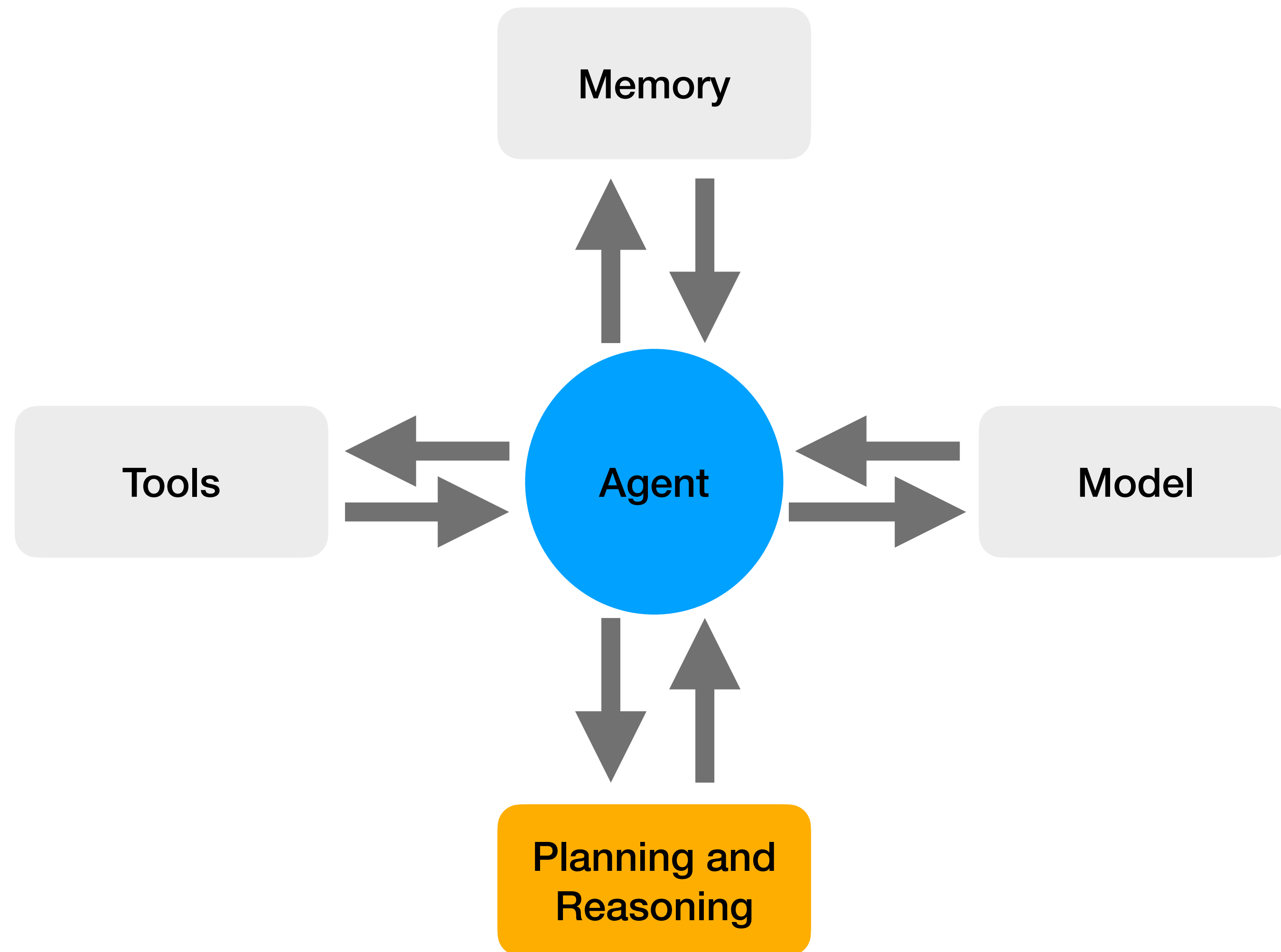
What is a Memory?

- LLMs are essentially stateless.
- Memory in AI agents is the ability to store, recall, and use past information (from conversations, interactions, or tasks) to influence future reasoning and actions.
- Provides context continuity across interactions.
- Enables personalization by remembering user preferences.
- Helps agents learn from past actions and refine behavior

Types of Memory



Let us finally focus on the planning and reasoning



Reasoning vs Planning

Reasoning

Problem decomposition and inference

Chain-of-thought logic construction

Pattern recognition and analysis

Understanding causal relationships

Reasoning = thinking through the problem

Planning

Action selection and sequencing

Resource allocation decisions

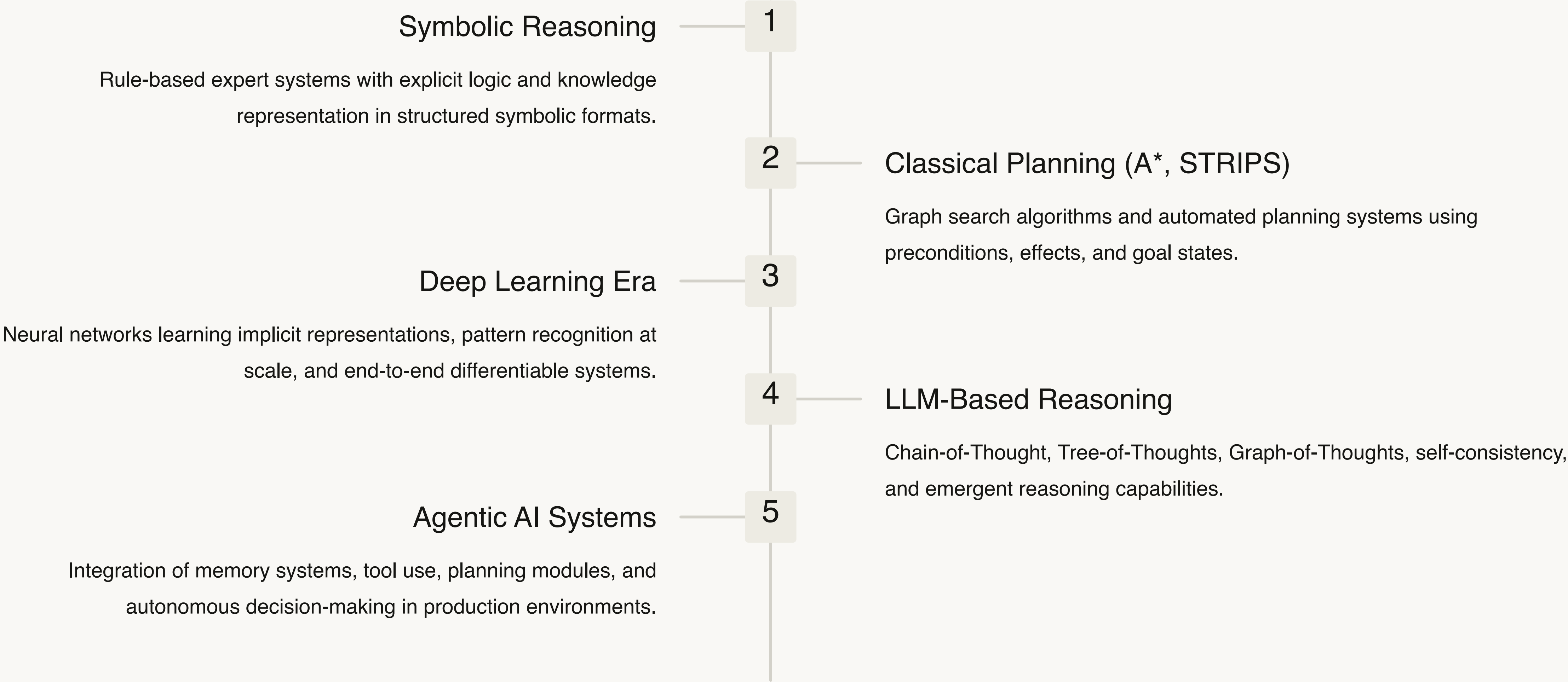
Temporal ordering of tasks

Execution path determination

Planning = organizing steps to execute

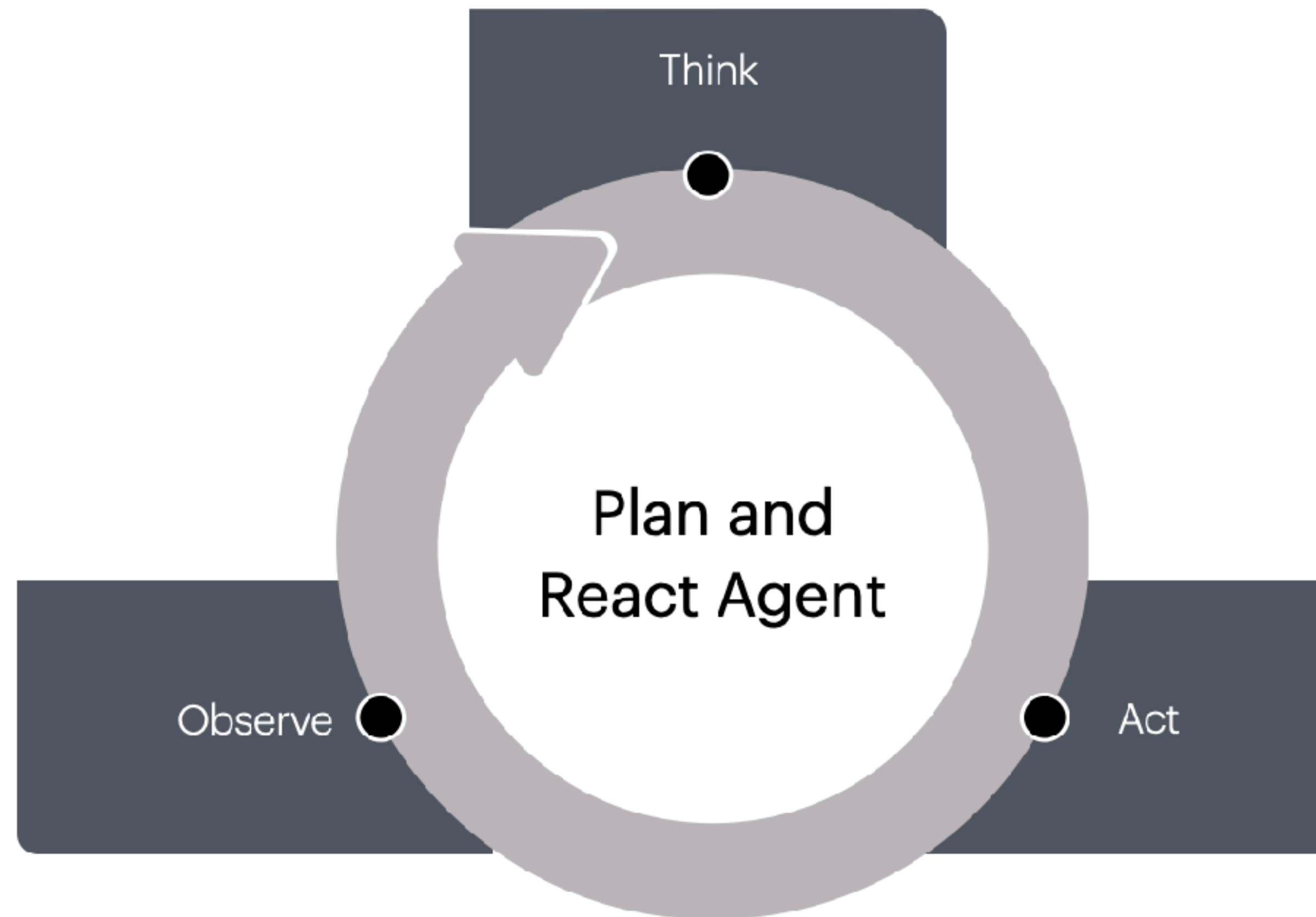
Core distinction: Reasoning generates insights; planning translates insights into actionable, ordered execution strategies.

Evolution of Reasoning

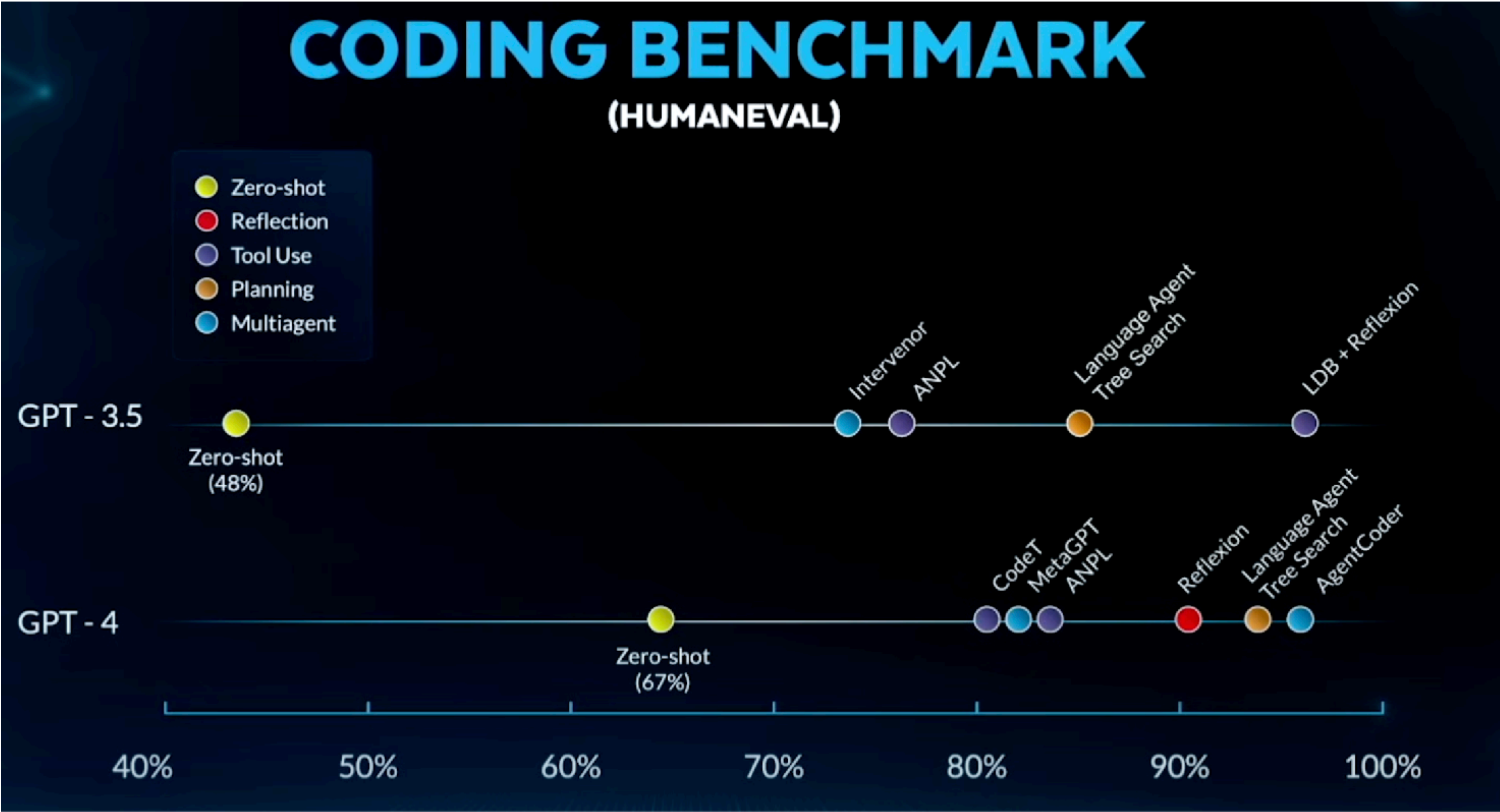


Why Planning and Reasoning?

- Agents enhance the performance of LLMs by iterating between planning and critiquing



Agents are amazing!



The promise of Agentic AI in 2026



"The age of agentic AI has arrived," according to Forbes.



"Agentic AI will Autonomously resolve 80% of common customer service issues without human intervention by 2029," according to a Gartner press release.



We believe that, in 2025, we may see the first AI agents 'join the workforce' and materially change the output of companies- Sam Altman



Nvidia's Jensen Huang says AI agents are 'a multi-trillion-dollar opportunity' and 'the age of AI Agentics is here'

Opportunities in Enterprises

Customer Agents

Deliver personalized customer experiences by answering questions, resolving issues, & making recommendations.

Employee Agents

Boost productivity by streamlining processes, managing repetitive tasks, answering questions, as well as editing and translating critical content and communications.

Creative Agents

Supercharge the design and creative process by generating content, images, and ideas, assisting with design, writing, personalization, and campaigns.

Data Agents

Built for complex data analysis. They have the potential to find and act on meaningful insights from data, all while ensuring the factual integrity of their results.

Coding Agents

Accelerate software development with AI-enabled code generation and coding assistance, and to ramp up on new languages and code bases.

Security Agents

Security agents strengthen security posture by mitigating attacks or increasing the speed of investigations.

Challenges of building and deploying agents in enterprise

POC Proliferation

Everyone's building proof-of-concepts, but few survive to become scalable products.

Non Determinism

LLM outputs are probabilistic and vary by prompt, context, or temperature. Enterprises need deterministic output

Immature Tools & Fragile Integrations

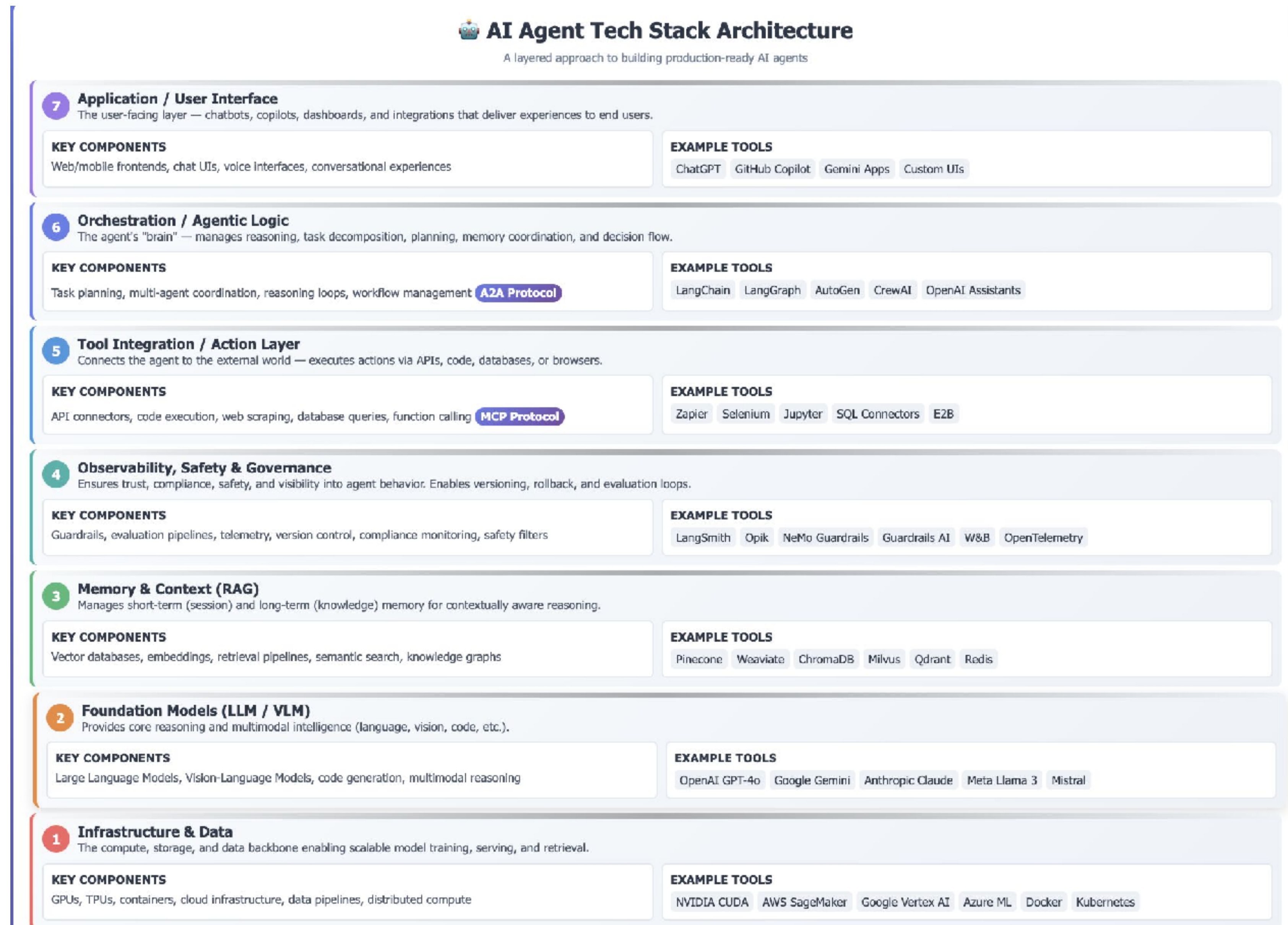
Tools are evolving but there are too many of them and often they are incompatible with one another

Skill Gap

Data scientists lack app dev muscle; engineers lack prompt/model intuition. Both lack business and language skills

AI Tech Stack & Tools Landscape

AI Tech Stack & Tools Landscape



Key takeaways and learning

- Agents
 - Components of agents
- Models
 - What are models , language models , and large language models
 - What do terms like context window, parameters and tokens mean
 - Types of models
- The opportunities to use agents in enterprise
- Challenges that enterprises need to address while using agents

Appendix

Reasoning Techniques in Agents

Chain-of-Thought (CoT)

Step-by-step reasoning where the model explicitly articulates intermediate logic steps before reaching conclusions.

Best for: Mathematical problems, multi-hop QA, logical deduction tasks

Example prompt: "Calculate the total cost of 3 items at \$15 each, plus a 10% tax. Show your work."

Self-Consistency

Generate multiple reasoning paths, then select the most frequent answer through majority voting across diverse solution trajectories.

Best for: Complex problems with multiple valid approaches, reducing variance

Example prompt: "If a and b are integers, and $a > b$, is $a \cdot b > 0$ possible? Explain your reasoning."

Tree-of-Thoughts (ToT)

Explores multiple reasoning branches simultaneously, evaluating and pruning paths based on intermediate state quality assessments.

Best for: Strategic planning, game playing, optimization problems

Example prompt: "Develop a strategy for winning a chess game from the current board state. Explore multiple lines of play."

Graph-of-Thoughts (GoT)

Models reasoning as a directed graph where thoughts can combine, split, and reference each other non-linearly.

Best for: Complex interdependent reasoning, synthesis tasks

Example prompt: "Analyze the multifaceted impact of AI on the job market, considering economic, social, and educational perspectives."

Reflection / Self-Critique

Agent evaluates its own outputs, identifies errors or gaps, and iteratively refines responses through self-assessment loops.

Best for: Code generation, content refinement, error correction

Example prompt: "Write a short story about a detective. Review your story for plot holes and improve the ending."

Planning Techniques in AI Agents

ReAct (Reason + Act)

Interleaves reasoning traces with action execution, allowing agents to dynamically adjust plans based on environmental feedback.

Plan-and-Execute

Generates complete action sequences upfront, then executes deterministically with error handling and rollback mechanisms.

Reflexion / Self-Refine

Incorporates execution feedback into memory, enabling agents to learn from failures and improve future planning iterations.

Hierarchical Planning

Meta-agent decomposes high-level goals into sub-tasks, delegating execution to specialized worker agents with domain expertise.

Task Decomposition Frameworks

DSPy, LangGraph, and AutoGen provide structured decomposition, graph-based execution, and multi-agent orchestration primitives.